The 2015 Ukraine Blackout: Implications for False Data Injection Attacks

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***Abstract*—In a false data injection attack (FDIA), an adversary stealthily compromises measurements from electricity grid sensors in a coordinated fashion, with a view to evading detection by the power system bad data detection module. A successful FDIA can cause the system operator to perform control actions that compromise either the physical or economic operation of the power system. In this letter, we consider some implications for FDIAs arising from the late 2015 Ukraine Blackout event.**

***Index Terms*—Cyber-attacks, false data injection attacks, ukraine blackout.**

I. Background on the 2015 Ukraine Blackout

On 23 December 2015, a synchronized and coordinated cyber-attack compromised three Ukrainian regional electric power distribution companies, resulting in power outages affecting approximately 225,000 customers for several hours [1]. This cyber-attack entailed several technical components, probably requiring extensive reconnaissance of the victim networks by adversaries [2]: variants of the BlackEnergy 3 malware were reportedly delivered via spear phishing emails and may have been used as an initial access vector for the theft of authorized users' virtual private network (VPN) credentials; a telephonic denial-of-service attack was executed to frustrate reports of outages to call centers; and a modified KillDisk firmware attack erased master boot records on workstations, thereby delaying restoration efforts.

These security compromises enabled the primary attack, namely a hijack of the Supervisory Control and Data Acquisition (SCADA) network, including the targeting of field devices with malicious firmware, thereby facilitating the remote opening of substation breakers. Manual operations were ultimately required to restore electrical service to customers [2].

Manuscript received June 30, 2016; accepted October 19, 2016. Date of publication November 22, 2016; date of current version June 16, 2017. This work supported in part by China Southern Power Grid under Grant WYKJ00000027. Paper no. PESL-00113-2016.

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Digital Object Identifier 10.1109/TPWRS.2016.2631891

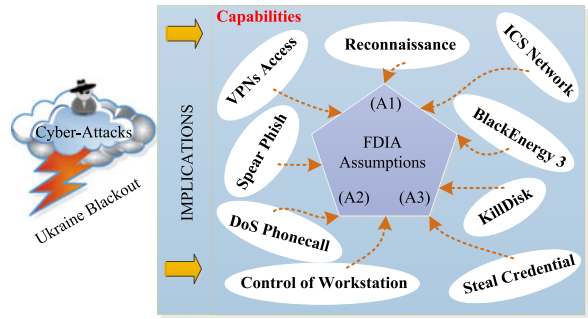

Fig. 1. Cyber-attackers' capabilities towards power systems.

II. Implications

A. Assumptions for FDIAs

The false data injection attack (FDIA), first proposed by Liu *et al.* [3], is a cyber-attack in which power system state estimation outputs are corrupted by injecting false data into meter measurements in a carefully coordinated fashion. The defining feature of a successful FDIA is that the state estimation residual falls below a hypothesis test threshold despite the presence of corrupted measurements, the attack thereby evading detection.

Research on FDIAs has been extensive [3]–[7], typically relying on three key assumptions [3], [4], namely that the attacker:

(A1) has knowledge of power system operations and features of the target system;
(A2) is capable of manipulating meter measurements; and
(A3) has knowledge of important control and operation information, such as the network topology, system electrical parameters, specifics of the SCADA network devices and bad data detection scheme, etc.

Considered jointly, (A1)–(A3) are strong assumptions, potentially calling into question the practical feasibility of mounting a successful FDIA. Nevertheless, we contend that the Ukraine blackout underscores the plausibility of assumptions (A1)–(A3), shown in Fig. 1, and that the "strongest capability of the [Ukraine blackout] attackers was [. . .] their capability to perform long-term reconnaissance operations required to learn

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the environment and execute a highly synchronized, multistage, multisite attack" [2].

B. Aspects Contributing to Feasibility of FDIAs

1) Ready Access to Information Sources: In the Ukraine blackout, adversaries showed expertise. Besides the intrusion of information technology (IT) networks, they were able to wrest control from substation control centers and operate the Human Machine Interface (HMI) in the supervisory control system to switch on breakers remotely.

The Internet provides potential adversaries with ready access to substantial quantities of high-quality, power system and vendor-specific information: textbooks introduce the foundations of generation, transmission, and distribution systems; research publications update scholars' knowledge and power system innovations; industrial control system (ICS) vendors readily share basic functionality and system architectures on web portals; and industrial control standards and network protocols are freely distributed on the Internet. Collectively, these resources provide determined adversaries with considerable knowledge-base with which to inform a cyber-attack.

2) Utilization of Vulnerabilities: Vulnerabilities in firewalls, network protocols, encryption, and VPN connections serve as "half-open-doors" to informed adversaries. There are three approaches to manipulating meter measurements for FDIAs: (i) compromising meters locally; (ii) intercepting and forging data packets when transferring to the control center; and (iii) modifying control center database [4].

In the Ukraine blackout, adversaries loaded malicious firmware into SCADA network field gateway devices to ensure that even when operator workstations were recovered, remote commands could not be issued to bring the substations back online [2]. Based on this, it can be foreseen that there would be high probabilistic for the adversaries to locally manipulate device parameters and gathered measurements by building direct access to field devices.

Furthermore, intercepting and forging communication messages is easy to achieve, since the SCADA layer communication network architecture has no cryptographically secure communication protocol.

In the Ukraine blackout, password-protected access to substation control center workstations was likely gained by adversaries via keystroke loggers [2]. The adversaries could exploit stolen credentials via VPNs to enter ICS network. Information stored on SCADA servers can be then readily stolen. Database manipulation is therefore possible after successfully gaining credentialed access.

3) Reconnaissance: The Ukraine cyber-attack likely followed long-term power system reconnaissance over six months or more without being noticed [2]. According to E-ISAC and SANS, long-term reconnaissance is considered to be the strongest capability required of attackers in launching this highly synchronized, multistage, multisite cyber-attack.

Knowledge of network topology and parameters required for a FDIA can be gained not only via one-off observations of components, but also from market data or measured power flows over extended periods. Additionally, system operators may also be proactively or passively tracked, decoyed, and deceived into divulging important configuration or operation information to adversaries. Moreover, research on FDIAs employing incomplete topology information and limited meter manipulation capability has been validated as feasible [8]. Assumptions (A2) and (A3) are therefore not unreasonable given extended periods of reconnaissance.

The 2015 Ukrainian blackout illustrates that sufficiently determined adversaries are able to operate for extended periods, progressively accumulating key operational information in a covert fashion. By extension, an attacker whose motivation is economic advantage could feasibly launch a successful FDIA by concealing their activities over an extended period.

C. Recommendations

Both ICS-CERT [1] and SANS [2] have provided recommendations and defense strategies regarding the Ukraine blackout incident. We further recommend that regular staff cyber-security training should be regarded as fundamental to power system security. Such training would strengthen defenses against FDIAs and coordinated cyber-attacks alike. Moreover, we recommend that components such as remote terminal units (RTUs), switches, breakers etc. should support both automatic and manual modes, in the event of failures in automatic restoration.

III. Conclusion

The 2015 Ukraine blackout is the first publicly acknowledged incident caused by a cyber-attack. The event highlights the vulnerability of highly automated, cyber information-based smart grid environments to coordinated cyber-attacks. In particular, this letter argues that the circumstances of the Ukraine blackout underscore the plausibility of common assumptions regarding the knowledge and capabilities required by an adversary in order to mount a successful FDIA on a power system.

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